AI 3642 Project Proposal

1. Group Members

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2. Project Idea

Title: Predicting Soccer Match Outcomes Using Machine Learning and Data Mining

Summary

For our project we want to try predicting soccer match outcomes (home win, away win, or draw) using stuff like goals scored, shots, and possession from past games. We're combining two of the areas from the project list, Machine Learning and Data Mining, so we can each own one piece: Khadim is going to work on grouping teams into tiers based on their stats (data mining), and Ryann is going to build the classifier that actually predicts the outcome (machine learning) using those groups plus the raw stats. We both like soccer, and this let us split the work into two clear pieces instead of both doing the same thing.

Preliminary Reading

We looked at a few papers to check that this was actually doable in one semester. Ren and Susnjak (2022) built classifiers for English Premier League matches and found that adding engineered features, including Elo-style ratings, helped, and that ensemble methods worked best out of everything they tried. Osman (2024) tried a handful of classifiers, including Random Forest and Logistic Regression, on Premier League data and got over 70% accuracy once they simplified it down to just win/loss. Beal et al. (2021) got around 63% accuracy using match stats plus some extra context over several Premier League seasons. Patel (2024) combined Random Forest, Decision Tree, and Gradient Boosting into one ensemble, did some feature selection and class balancing, and ended up with a very high accuracy number on their processed dataset. Reading these made us feel a lot better about pulling this off in a semester, and gave us the idea to split the project into the two parts below.

Proposed Methodology

1. **Data collection and cleaning:** grab historical match records (teams, date, goals, shots, possession, standings) and clean/format them so we can actually use them.

2. **Data mining – team clustering (Khadim):** run a clustering method (probably k-means) on rolling team stats to group teams into performance tiers or playing-style clusters. This is basically the same idea as the Elo-style feature engineering in Ren and Susnjak (2022), and turns raw stats into one feature that sums up how strong a team is.
3. **Machine learning – outcome classifier (Ryann):** train a classifier (starting with Logistic Regression and/or Random Forest) using the cluster feature plus the raw match stats to predict win, loss, or draw.
4. **Evaluation:** check accuracy and a confusion matrix, and compare our results to a simple baseline (always guess home win) and to the accuracy ranges from the papers above.
5. **Stretch goal (if we have time):** a small GUI or notebook demo where you pick two teams and it shows you the predicted outcome.

Data Source

We're planning on using the "European Soccer Database" from Kaggle, since it already has match results plus team and player stats for a bunch of European leagues from 2008 to 2016. If that doesn't have everything we need, we'll grab extra match stats from football-data.co.uk too, since it's free and easy to download.

References

Beal, R., Middleton, S. E., Norman, T. J., & Ramchurn, S. D. (2021). Combining machine learning and human experts to predict match outcomes in football: A baseline model. Proceedings of the AAAI Conference on Artificial Intelligence, 35(17), 15447–15451.

Osman, O. (2024). Predicting football matches outcomes using machine learning approaches [Master's dissertation, Hamad Bin Khalifa University].

Patel, D. (2024). Predicting football match outcomes with ensemble machine learning models [Master's dissertation, Dublin Business School].

Ren, Y., & Susnjak, T. (2022). Predicting football match outcomes with explainable machine learning and the Kelly index. arXiv:2211.15734.